

Pre-Characterization Process Risk Assessment

A-Mab Drug Substance — Risk Ranking and Study-Type Assignment — RA-001

Manufacturing Science & Technology (MSAT) / Quality Risk Management

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Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Quality Risk Management / MSAT	—	—	—
Reviewer	Process Development	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

CPP, critical process parameter · CQA, critical quality attribute · D, detection · DoE, design of experiments · DS, drug substance · FMEA, failure mode and effects analysis · GPP, general process parameter · ICH, International Council for Harmonisation · KPP, key process parameter · MSAT, Manufacturing Science and Technology · O, occurrence · PC, process characterization · RPN, risk priority number · RRF, risk ranking and filtering · S, severity · WC-CPP, well-controlled CPP.

1 Purpose and scope

This Pre-Characterization Process Risk Assessment (**RA-001**) prioritizes the A-Mab drug-substance process parameters for characterization and assigns each parameter a study type, **before** the characterization studies are executed. It is the first (pre-hoc) risk assessment of the A-Mab lifecycle of iterative risk assessments: it uses prior knowledge and early development experience — not the results of the characterization studies — to decide which parameters carry a credible risk to a critical quality attribute (CQA) and therefore warrant a multivariate designed experiment, which warrant a univariate study, and which require only routine control. Its output is the **characterization scope** that is handed to each per-unit-operation Process Characterization Plan (**PCP-003 ... PCP-010**).

Scope. The 37 process parameters of the drug-substance train (production bioreactor through final UF/DF), assessed for their prospective (pre-characterization) risk of impacting the 10 drug-substance CQAs and for the resulting study-type assignment. This assessment does **not** classify parameters as CPP/WC-CPP/KPP/GPP — that is an output of the characterization studies, reported in each PCR and consolidated in the Process Characterization Master Report (**PCMR-001**); nor does it use the post-characterization residual risk.

Basis. The assessment is derived from the Process Transfer Plan (**PTP-001**) product and process description and follows the A-Mab quality-risk-management methodology (CMC Biotech Working Group 2009) — the two-axis CQA criticality tools and the risk-ranking-and-filtering (RRF) study-type rule — within the ICH Q9 quality-risk-management framework (International Council for Harmonisation 2023) and the QbD approach of ICH Q8 and Q11 (International Council for Harmonisation 2009, 2012). The curated per-parameter failure-mode and effect content is the same content source used to build the post-characterization process FMEA.

2 Related documents

This assessment is a parent risk document for the characterization package; the full package is cross-referenced for traceability.

Table 2.1: The A-Mab drug-substance document package referenced by this risk assessment.

Document ID	Document class	Subject
PTP-001	Process Transfer Plan	A-Mab Drug Substance
PCMP-001	Process Characterization Master Plan	A-Mab Drug Substance
PCP-003	Process Characterization Plan	Production Bioreactor (Step 3)
PCR-003	Process Characterization Report	Production Bioreactor (Step 3)
PCP-004	Process Characterization Plan	Harvest and Clarification (Step 4)
PCR-004	Process Characterization Report	Harvest and Clarification (Step 4)
PCP-005	Process Characterization Plan	Protein A Chromatography (Step 5)
PCR-005	Process Characterization Report	Protein A Chromatography (Step 5)
PCP-006	Process Characterization Plan	Low-pH Viral Inactivation (Step 6)
PCR-006	Process Characterization Report	Low-pH Viral Inactivation (Step 6)
PCP-007	Process Characterization Plan	Cation Exchange Chromatography (Step 7)
PCR-007	Process Characterization Report	Cation Exchange Chromatography (Step 7)
PCP-008	Process Characterization Plan	Anion Exchange Chromatography (Step 8)
PCR-008	Process Characterization Report	Anion Exchange Chromatography (Step 8)
PCP-009	Process Characterization Plan	Small-Virus Retentive Filtration (Step 9)
PCR-009	Process Characterization Report	Small-Virus Retentive Filtration (Step 9)
PCP-010	Process Characterization Plan	Ultrafiltration / Diafiltration (Step 10)
PCR-010	Process Characterization Report	Ultrafiltration / Diafiltration (Step 10)
PCMR-001	Process Characterization Master Report	A-Mab Drug Substance

3 Risk-assessment methodology

3.1 Two distinct criticality assessments

Consistent with the A-Mab case study, two distinct assessments are kept separate (CMC Biotech Working Group 2009). The **CQA criticality** (quality-attribute criticality) is scored on two axes only — Tool #1: Risk Score = Impact $\times$ Uncertainty, and Tool #2: Severity $\times$ Likelihood — explicitly excluding manufacturing capability and detectability; this yields a continuum of criticality, not a binary critical / non-critical split. The **process-parameter risk** (this assessment) then asks, for each parameter, whether its variability could impact a CQA and how the parameter should therefore be studied. Criticality is treated as a continuum throughout (ICH Q9) (International Council for Harmonisation 2023).

3.2 Pre-characterization risk ranking and study-type assignment

Because the characterization studies have not yet been executed, this assessment uses the **initial** (pre-characterization) risk: a parameter with a credible impact to a CQA has an uncertain effect and poor in-process detectability (the effect is often detectable only at a downstream step or at release), so its initial risk is high and it is prioritized for study. A risk-ranking-and-filtering (RRF) screen scores each parameter’s potential main-effect and interaction impact and assigns the study type accordingly: a parameter with a credible main-effect and interaction risk to a CQA is carried into the unit operation’s **multivariate designed experiment (DoE)**; a parameter with a credible CQA impact but no expected interaction is studied by a **justified univariate** study; and a parameter with no credible CQA impact — one that affects only process performance — is studied **univariately** for its performance effect (CMC Biotech Working Group 2009). The classification of each parameter (CPP / WC-CPP / KPP / GPP) is deliberately **not** assigned here; it is an output of the executed studies.

3.3 Prospective severity and initial risk

The prospective severity of each parameter is taken as the maximum severity of the CQA(s) it could impact, on the A-Mab severity scale (2–10). The initial risk priority number (RPN = Severity $\times$ Occurrence $\times$ Detection) uses the pre-characterization occurrence and detection: an uncertain effect (higher occurrence) and poor pre-characterization detectability (highest for viral-clearance attributes, which cannot be detected by in-process testing). The initial RPN is used only to rank and prioritize; it is expected to fall after characterization and the resulting control strategy, as shown in the post-characterization process FMEA.

4 CQA criticality framework

The CQAs in scope and their criticality (Tool #1 = Impact $\times$ Uncertainty; Tool #2 severity) are given in Table 4.1. Attributes rated High (H) or Very High (VH) are treated as *Critical*, and their associated parameters carry the highest prospective severity.

Table 4.1: Drug-substance CQA criticality framework (Tool #1 = Impact $\times$ Uncertainty; Tool #2 = severity).

CQA	Category	Acceptance	Criticality	Tool #1	Tool #2 sev.
Afucosylation	glycosylation	2–13 % of glycans	H	60	7
Galactosylation (total %Gal)	glycosylation	10–40 % (G1+G2 equiv.)	H	48	7
High Mannose	glycosylation	3–10 % of glycans	VH	80	7
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60	5
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4	3
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36	5
Residual DNA	process impurity	0–0.001 ng/dose	VL	6	1
Leached Protein A	process impurity	0–5 ppm	L-M	16	5
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20	9
Viral clearance — MVM (parvovirus)	viral safety	8.6–20 log10 (cumulative)	VH	20	9

5 Process-parameter risk ranking and study-type assignment

The prospective risk ranking of every process parameter, with the CQA(s) at risk, the prospective severity, the initial (pre-characterization) RPN, the assigned study type and the characterization priority, is given in Table 5.1. Parameters that could impact a CQA are prioritized for study; parameters that affect only process performance are ranked Low (product-quality) priority and studied univariately for their performance effect. The prospective failure mode and effect underlying each ranking are tabulated in Appendix A.

Table 5.1: Pre-characterization process-parameter risk ranking and study-type assignment (initial RPN = prospective Severity $\times$ pre-characterization Occurrence $\times$ Detection).

Step	Unit operation	Parameter	CQA(s) at risk	Sev.	Init. RPN	Study type	Pri-or-ity
3	Production Bioreactor	Culture pH	Afucosylation, Galactosylation (total %Gal), High Mannose, Acidic charge variants (deamidation), Aggregates (HMW)	10	560	multi-variate DoE	High
3	Production Bioreactor	Culture temperature	Afucosylation, Galactosylation (total %Gal), Acidic charge variants (deamidation)	8	448	multi-variate DoE	High
3	Production Bioreactor	Dissolved CO2 (pCO2)	Galactosylation (total %Gal), Acidic charge variants (deamidation)	8	448	multi-variate DoE	High
3	Production Bioreactor	Osmolality	Galactosylation (total %Gal), Acidic charge variants (deamidation)	8	448	multi-variate DoE	High
3	Production Bioreactor	Culture duration	Afucosylation, Galactosylation (total %Gal), Aggregates (HMW), Host Cell Protein (HCP), Residual DNA	8	448	multi-variate DoE	High
3	Production Bioreactor	Dissolved oxygen	process performance	4	96	univariate	Low
3	Production Bioreactor	Initial viable cell conc.	process performance	4	96	univariate	Low
3	Production Bioreactor	Basal medium concentration	process performance	4	96	univariate	Low

Step	Unit operation	Parameter	CQA(s) at risk	Sev.	Init. RPN	Study type	Pri- or- ity
3	Production Bioreactor	Nutrient feed-1 volume	process performance	4	96	univariate	Low
4	Harvest and Clarification	Centrifugation (rcf)	process performance	4	96	univariate	Low
4	Harvest and Clarification	Depth filter load	process performance	4	96	univariate	Low
4	Harvest and Clarification	Post-clarification turbidity	process performance	4	96	univariate	Low
5	Protein A Chromatography	Protein load	Host Cell Protein (HCP)	6	336	multivariate DoE	Medium
5	Protein A Chromatography	Elution buffer pH	Host Cell Protein (HCP)	6	336	multivariate DoE	Medium
5	Protein A Chromatography	Load flow rate	process performance	4	96	multivariate DoE	Low
5	Protein A Chromatography	End of pool collect	process performance	4	96	multivariate DoE	Low
5	Protein A Chromatography	Operating temperature	process performance	4	96	univariate	Low
5	Protein A Chromatography	Bed height	process performance	4	96	univariate	Low
6	Low-pH Viral Inactivation	Inactivation pH	Viral clearance — XMuLV (enveloped), Aggregates (HMW)	10	700	multivariate DoE	High
6	Low-pH Viral Inactivation	Hold time	Aggregates (HMW), Viral clearance — XMuLV (enveloped)	10	700	multivariate DoE	High

Step	Unit operation	Parameter	CQA(s) at risk	Sev.	Init. RPN	Study type	Pri- or- ity
6	Low-pH Viral Inactivation	Temperature	Viral clearance — XMuLV (enveloped)	10	700	multi-variate DoE	High
6	Low-pH Viral Inactivation	A-Mab concentration	process performance	4	96	univariate	Low
7	Cation Exchange Chromatography	Protein load	Aggregates (HMW), Host Cell Protein (HCP)	8	448	multi-variate DoE	High
7	Cation Exchange Chromatography	Load/Wash conductivity	Host Cell Protein (HCP)	6	336	multi-variate DoE	Medium
7	Cation Exchange Chromatography	Elution buffer pH	Aggregates (HMW)	8	448	multi-variate DoE	High
7	Cation Exchange Chromatography	Elution stop collect	Aggregates (HMW)	8	448	multi-variate DoE	High
7	Cation Exchange Chromatography	Elution flow rate	process performance	4	96	univariate	Low
8	Anion Exchange Chromatography	Load pH	Host Cell Protein (HCP), Viral clearance — XMuLV (enveloped), Viral clearance — MVM (parvovirus)	10	700	multi-variate DoE	High
8	Anion Exchange Chromatography	Equil/Wash-1 conductivity	Host Cell Protein (HCP)	6	336	multi-variate DoE	Medium

Step	Unit operation	Parameter	CQA(s) at risk	Sev.	Init. RPN	Study type	Pri- or- ity
8	Anion Exchange Chromatography	Load conductivity	Viral clearance — XMuLV (enveloped), Viral clearance — MVM (parvovirus)	10	700	multi-variate DoE	High
8	Anion Exchange Chromatography	Protein load	Host Cell Protein (HCP)	6	336	multi-variate DoE	Medium
8	Anion Exchange Chromatography	Operating flow rate	Viral clearance — XMuLV (enveloped), Viral clearance — MVM (parvovirus)	10	700	univariate (justified)	High
9	Small-Virus Retentive Filtration	Filtration volume (load)	Viral clearance — MVM (parvovirus)	10	700	multi-variate DoE	High
9	Small-Virus Retentive Filtration	Filtration pressure	Viral clearance — MVM (parvovirus), Viral clearance — XMuLV (enveloped)	10	700	multi-variate DoE	High
10	Ultrafiltration / Diafiltration	Number of diavolumes	process performance	4	96	univariate	Low
10	Ultrafiltration / Diafiltration	Transmembrane pressure	process performance	4	96	univariate	Low
10	Ultrafiltration / Diafiltration	Final DS concentration	process performance	4	96	univariate	Low

Of the 37 parameters, 21 carry a credible prospective risk to a CQA and 16 affect only process performance. 16 are ranked **High** priority (a CQA of severity 8 at risk), 5 **Medium**, and 16 **Low**.

6 Characterization scope

The study-type assignment defines the characterization scope handed to the per-unit-operation plans (Table 6.1): 22 parameters are assigned to a **multivariate DoE**, 1 to a **justified univariate** study, and 15 to **univariate** ranging. Multivariate designed experiments are therefore planned for 6 unit operations (Steps 3, 5, 6, 7, 8, 9); the remaining steps — harvest / clarification and final UF/DF — carry no credible CQA risk and are characterized univariately.

Table 6.1: Characterization scope handed to the Process Characterization Plans (PCP-003 ... PCP-010).

Assigned study type	Parameters	RRF basis
Multivariate DoE	22	Credible main-effect and interaction risk to a CQA
Univariate (justified)	1	Credible CQA risk without an expected interaction
Univariate	15	No credible CQA risk; process-performance study only

This scope is carried into each Process Characterization Plan, which specifies the exact designs, ranges and acceptance criteria; the executed studies then classify each parameter and define the design spaces and control strategy (reported in the PCRs and consolidated in **PCMR-001** and the post-characterization process FMEA).

7 Roles and responsibilities

Table 7.1: Roles and responsibilities.

Function	Responsibility
Quality Risk Management / MSAT	Author the assessment; run the RRF screen; assign study types
Process Development	Provide prior-knowledge risk input; execute the assigned studies under the PCPs
Analytical Development	Confirm the CQA criticality and the detectability inputs
Quality Assurance	Approve the assessment and the resulting characterization scope

8 Approvals

Approval of this assessment authorizes the characterization scope and study-type assignments carried into the Process Characterization Plans. Signatures are recorded in Table [1](#).

References

- CMC Biotech Working Group. 2009. *A-Mab: A Case Study in Bioprocess Development*. CASSS.
- International Council for Harmonisation. 2009. *ICH Q8(R2): Pharmaceutical Development*. ICH.
- International Council for Harmonisation. 2012. *ICH Q11: Development and Manufacture of Drug Substances*. ICH.
- International Council for Harmonisation. 2023. *ICH Q9(R1): Quality Risk Management*. ICH.

Appendix A — Prospective failure modes and effects

The prospective (pre-characterization) failure mode and effect underlying each parameter’s risk ranking, from the curated A-Mab failure-mode content source. These are the hypotheses the characterization studies test; the confirmed effects and the resulting classification are reported in the PCRs and the post-characterization process FMEA.

Table 8.1: Appendix A — prospective failure modes and effects per process parameter.

Step	Parameter	Potential failure mode	Potential effect	CQA(s) at risk
3	Culture pH	pH excursion outside the characterized range (6.6-7.1)	Shifts glycosylation (ADCC via afucosylation; CDC via galactosylation) and charge/aggregate profile	Afucosylation, Galactosylation (total %Gal), High Mannose, Acidic charge variants (deamidation), Aggregates (HMW)
3	Culture temperature	Temperature excursion outside 34-36 C	Alters afucosylation, galactosylation and acidic charge variants; interacts with pH	Afucosylation, Galactosylation (total %Gal), Acidic charge variants (deamidation)
3	Dis-solved CO2 (pCO2)	Dissolved CO2 outside 40-160 mmHg	Strong effect on galactosylation and acidic variants	Galactosylation (total %Gal), Acidic charge variants (deamidation)
3	Osmolality	Osmolality outside 360-440 mOsm	Affects galactosylation and acidic variants	Galactosylation (total %Gal), Acidic charge variants (deamidation)
3	Culture duration	Culture extended beyond 15-19 days / harvested on low viability	Lowers afucosylation & galactosylation; raises HCP/DNA and aggregate as viability declines	Afucosylation, Galactosylation (total %Gal), Aggregates (HMW), Host Cell Protein (HCP), Residual DNA
3	Dis-solved oxygen	DO outside 30-70%	Impacts peak VCD, titer and process consistency (no significant CQA impact)	process performance
3	Initial viable cell conc.	Inoculation density outside 0.5-1.5e6/mL	Drives peak/integral VCD and titer; no product-quality impact	process performance

Step	Parameter	Potential failure mode	Potential effect	CQA(s) at risk
3	Basal medium concentration	Basal medium concentration outside 0.8-1.6X	Statistically significant but shallow effect on afucosylation; drives titer	process performance
3	Nutrient feed-1 volume	Nutrient feed-1 volume outside 9-15% WV	Slight effect on galactosylation; critical for titer	process performance
4	Centrifugation (rcf)	Centrifugation rcf outside 6000-12000 g	Affects clarification/yield (no CQA impact)	process performance
4	Depth filter load	Depth filter load outside 80-160 L/m ²	Affects filter capacity/turbidity (no CQA impact)	process performance
4	Post-clarification turbidity	Post-clarification turbidity high	Feed quality to Protein A (no CQA impact)	process performance
5	Protein load	Protein load above characterized range (>50 g/L resin)	Increases pool HCP (interaction with elution pH)	Host Cell Protein (HCP)
5	Elution buffer pH	Elution pH below characterized range (<3.2)	Increases pool HCP; also sets up viral inactivation feed	Host Cell Protein (HCP)
5	Load flow rate	Load flow rate high with high load	Reduces product yield (no CQA impact)	process performance
5	End of pool collect	End-of-pool collection outside 2.0-3.2 CV	Affects product yield (no CQA impact)	process performance
5	Operating temperature	Parameter outside characterized range	No significant product-quality impact expected	process performance
5	Bed height	Parameter outside characterized range	No significant product-quality impact expected	process performance
6	Inactivation pH	pH > 4.0 (incomplete inactivation) or pH < 3.2 (aggregation/precipitation)	Loss of XMuLV inactivation assurance or increased aggregate	Viral clearance — XMuLV (enveloped), Aggregates (HMW)

Step	Parameter	Potential failure mode	Potential effect	CQA(s) at risk
6	Hold time	Hold time < 60 min (incomplete) or > 180 min (aggregation)	Incomplete inactivation or increased aggregate	Aggregates (HMW), Viral clearance — XMuLV (enveloped)
6	Temperature	Temperature < 15 C	Lower inactivation rate at short time / higher pH	Viral clearance — XMuLV (enveloped)
6	A-Mab concentration	Concentration > 35 g/L	No significant effect on inactivation, aggregation or charge variants	process performance
7	Protein load	Protein load above characterized range (>40 g/L resin)	Reduced aggregate and HCP clearance (interactions with pH and wash conductivity)	Aggregates (HMW), Host Cell Protein (HCP)
7	Load/Wash conductivity	Load/wash conductivity below 3 mS/cm	Reduced HCP clearance	Host Cell Protein (HCP)
7	Elution buffer pH	Elution pH outside 5.8-6.2	Alters aggregate distribution and yield	Aggregates (HMW)
7	Elution stop collect	Extended elution stop-collect (>1.5 OD)	Minor increase in pool aggregate	Aggregates (HMW)
7	Elution flow rate	Elution flow rate outside 100-300 cm/hr	Affects peak shape and yield (minor aggregate)	process performance
8	Load pH	Load pH below characterized range (<7.2)	Reduced HCP removal and reduced XMuLV/MVM clearance	Host Cell Protein (HCP), Viral clearance — XMuLV (enveloped), Viral clearance — MVM (parvovirus)
8	Equil/Wash-1 conductivity	Equil/Wash-1 conductivity above 3.6 mS/cm	Reduced HCP removal	Host Cell Protein (HCP)
8	Load conductivity	Load conductivity above characterized range	Reduced viral clearance (interacts with pH)	Viral clearance — XMuLV (enveloped), Viral clearance — MVM (parvovirus)
8	Protein load	Protein load above characterized range (>300 g/L resin)	Reduced HCP removal capacity (viral clearance robust)	Host Cell Protein (HCP)
8	Operating flow rate	Operating flow rate above 450 cm/hr	Reduced residence time / viral clearance margin	Viral clearance — XMuLV (enveloped), Viral clearance — MVM (parvovirus)

Step	Parameter	Potential failure mode	Potential effect	CQA(s) at risk
9	Filtration volume (load)	Volumetric load above 105 L/m ²	Reduced MVM log-reduction (parvovirus breakthrough risk)	Viral clearance — MVM (parvovirus)
9	Filtration pressure	Pressure outside filter operating limits	Potential impact on virus removal at high load	Viral clearance — MVM (parvovirus), Viral clearance — XMuLV (enveloped)
10	Number of diavolumes	Diavolumes < 7	Incomplete buffer exchange (formulation; not part of DS characterization)	process performance
10	Transmembrane pressure	TMP outside 10-15 psi	Affects flux/processing (formulation)	process performance
10	Final DS concentration	Final concentration outside 65-85 g/L	DS concentration spec (formulation)	process performance